Gabriel Carvalho Silva

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Spoken Languages: English (Fluent, C2 Level), Portuguese (Native)

Professional Summary

Software Engineer with 5 years of experience developing high-performance solutions. Prooven experience in Java and Kotlin ecosystems, Microservices, and Legacy Modernization. Proven track record of bridging the gap between complex technical implementation and business requirements, ensuring resilience, security, and operational excellence in fast-paced environments.

Technical Skills

Languages: Kotlin, Java, Python, SQL, Shell Scripting, Golang

Architecture & Design: Microservices, Domain-Driven Design (DDD), Event-Driven Architecture, SOLID, Database Evolution, REST API Design

Frameworks & Testing: Spring Boot, Dropwizard, Hibernate/JPA, JUnit, Mockito, FastAPI

Infra & Observability: AWS, Kafka, MQTT, Docker, Kubernetes, AWS, Grafana, CI/CD Pipelines (e.g. GitHub Actions and Gitlab CI/CD)

Databases: PostgreSQL, MySQL, ScyllaDB, Redis, Flyway

Professional Experience

Software Engineer Mar 2026 – Present
ClassPass - Remote

- **Architecture & Legacy Modernization:** Spearheaded the refactoring of legacy flows to support multi-tenant architectures, enabling the seamless reuse of core business logic across diverse data shapes within a Kotlin microservices ecosystem.
- **Event-Driven Systems:** Architected and implemented complex asynchronous workflows using Event-Driven Architecture (AWS SQS/SNS), delivering a highly configurable cancellation policy engine (handling both user and venue rules) and pass-based booking resolutions.
- **Domain-Driven Design (DDD):** Restructured Protobuf service contracts applying DDD and the Open-Closed Principle, significantly improving cross-team maintainability, extensibility, and interface clarity.
- **Data Integration:** Engineered robust tracking and Business Intelligence (BI) integration pipelines, effectively bridging transactional microservices with analytical data layers.
- **Technical Leadership & Culture:** Accelerated ramp-up time by 50% to take ownership of end-to-end technical designs and squad coordination. Fostered engineering excellence by leading design discussions, driving bug bashes, and organizing technical book clubs.
- **Tech Stack:** Kotlin, Dropwizard, PostgreSQL, AWS (SQS/SNS), Protobuf, Microservices

Application Engineer Oct 2024 – Mar 2026
Foretellix - Remote

- **Lead Technical Integration:** Orchestrated the design and deployment of custom data integration infrastructures for enterprise clients (Toyota, Nvidia), translating high-level business requirements into technical architectural specifications.
- **Cloud Infrastructure Architecture:** Designed scalable execution environments on AWS, integrating industry-standard platforms (e.g., Carla) with proprietary software to ensure high availability.
- **Automation & Tooling:** Developed a custom OpenDRIVE (.xodr) map manipulation API (DOM-like structure) and automation scripts, accelerating map editing workflows and enabling programmatic querying of map elements.
- **Cross-Functional Collaboration:** Acted as the primary technical liaison between internal engineering teams and external business stakeholders, conducting impact assessments and aligning technical delivery with client timelines.
- **Tech Stack:** ASAM OpenScenario 2, C++, Python, Shell, AWS, React, Kotlin

Software Engineer Intern Feb 2021 – Jan 2023
iFood - Financial Services Tribe | Remote

- **Legacy Modernization:** Contributed to the architectural transformation of legacy payment microservices into modern, resilient Kotlin/Spring Boot applications, enhancing scalability without disrupting business operations.
- **Resilient Architecture & Scale:** Designed fault-tolerant, event-driven financial systems using Kafka, ensuring data consistency and processing millions of daily transactions with low latency.
- **Database Evolution:** Managed database evolution strategies using Flyway and PostgreSQL/ScyllaDB, ensuring zero-downtime migrations for high-volume datasets.
- **Internal Tooling & Standards:** Developed a shared error-handling library (DSL) adopted across the platform to standardize user feedback, conducting code reviews to ensure adoption and maintainability.
- **Observability:** Integrated observability tools (Datadog/Grafana) to monitor system health, significantly reducing mean-time-to-resolution (MTTR) for payment failures.
- **Tech Stack:** Kotlin, Java, Spring Boot, Kafka, AWS, PostgreSQL, ScyllaDB, Redis, Go (Golang)

Key Projects

Bachelor Thesis - IoT Smart Scale Architecture Aug 2025 – Dec 2025
USP | Hybrid

- **End-to-End System Design:** Architected a complete IoT solution applying Domain-Driven Design (DDD) and Hexagonal Architecture principles to ensure decoupling and testability.
- **Implementation:** Developed an MQTT broker for real-time device communication, integrated with a Spring Boot/Kotlin backend, SQL database, and embedded face recognition using TensorFlow Lite models.
- **Tech Stack:** ESP32, Mosquitto (MQTT), Kotlin, PostgreSQL, Spring Boot, Docker, React, TensorFlow

Education & Leadership

Universidade de São Paulo (USP) Mar 2020 – Dec 2025
Bachelor of Computer Science

Technische Universität Braunschweig - Germany Mar 2023 – Mar 2024
Exchange Student

Leadership

- Octobit Junior - Head of Projects (2024-2025)
- Computer Science Week - Head Director (2020-2021)

Certifications & References

Certifications: TOEFL iBT Score: 113/120 | High School Programming Competition: 2nd Place

References:

- Mr. Cristiano Hahn - Senior Software Engineer @ iFood - cristianohahn.contato@gmail.com
- Mr. Danilo de Luca - Senior Software Engineer @ Classpass - https://www.linkedin.com/in/danilopereiradeluca
- Mr. Mohamed Fasil Ahamed - Application Engineering Manager @ Foretellix - https://www.linkedin.com/in/mohamed-fasil-syed-ahamed-7a146789/
- Mr. Gianmarco Macaro - Application Engineering Director @ Foretellix - https://www.linkedin.com/in/gianmarco-macaro-a213367b/